says Dr Harish Shetty, a psychiatrist. "There are other alternative therapies as well, but writing is certainly emerging as a powerful tool to counter both emotional and cognitive disturbances."

LEARNING BETTER

What makes writing so effective? For one, it is said to slow down one's mind. The average typing speed is around 40 words per minute; by contrast, an average adult can write about 13 words per minute. This extra time, say experts, helps the mind pace its thoughts, absorb information, draw deeper conclusions and process emotions. There are also fewer distractions when you are writing by hand, hence improving your concentration. Dr P.R. Krishnan, senior neurologist at Fortis Hospitals, Bengaluru, recommends that writing by hand be taught early in life. "Or it is a missed opportunity," he adds.



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DR P.R. KRISHNAN
Senior neurologist, Fortis Hospitals, Bengaluru



"WRITING IS EMERGING AS A POWERFUL TOOL TO COUNTER EMOTIONAL AND COGNITIVE DISTURBANCES"

DR HARISH SHETTY
Psychologist

Experiments have shown that writing can change the way your brain functions. When writing, you are not only activating the motor cortex, which makes your hand move, but also the motor planning parts of the visual cortex to visualise the letters in your mind, not to mention the neural networks associated with reading and spelling. "When your hand moves, it activates large areas of the brain responsible for working memory, hand-eye coordination, language and reasoning. It can also stimulate the hippocampus to create long-term memory," says Dr Krishnan. Sanjana Chatlani, 32, who founded The Bombay Lettering Company, a calligraphy and design studio in Mumbai, says calligraphy has helped improve not only her emotional health but also her productivity. "I feel I am more focused and attentive to detail," she says.

There are other reasons to recommend writing by hand. A study from the Johns Hopkins University published in 2021 notes that writing can improve already refined motor skills, which appears to help adults learn general literacy-related skills faster than if they tried to learn the same material by typing or watching videos. These findings were built upon a 2021 study by the Norwegian University of Science and Technology, which used an electroencephalogram (EEG) to track brain activity during classroom learning. The study found that cursive handwriting produces the best learning and memory, even when using digital pens to write on a computer screen. "The use of pen and paper gives the brain more hooks to hang your memories on. Writing by hand creates much more activity in the sensorimotor parts of the brain," author Audrey van der Meer said in a 2020 news release in *Science Daily*. "A lot of senses are activated by pressing the pen on paper, seeing the letters you write, and hearing the sound you make while writing. These sensory experiences create contact between different parts of the brain and open it up for learning," she said.

Experts say that what differentiates people who appear to have higher intelligence is the constant exercising of their neural pathways. "The brain is like your hand, if you stop using it, it'll weaken. Writing benefits you anyway, but if you do unfamiliar or new writing tasks, they will strengthen your neural networks," says Dr Krishnan. Thus, learning to write in a new language or a new script, or even with a different hand, can have cognitive benefits. And perhaps calligraphy more than anything else. Since the practice involves drawing shapes and forming a script, and, at the same time, creating muscle memory to recreate the shapes, different parts of the brain get exercised. "Calligraphy is an art," says Chatlani. "On a digital screen, the machine is aware of only certain rules. But if I want to tweak a letter, add a swirl or a flourish, only handwritten script will allow that flow of expression."

Writing can be a means of release, a calming exercise when you are emotionally fraught, an outlet for creativity, or an expression of love. The effort one makes to write to someone makes receiving a handwritten note almost magical. "We are used to things being convenient. To take that extra step [to write] for someone leaves them with the feeling that you care enough for them," says Kejriwal. Dubey agrees. "Handwritten communication is something you cherish forever, and every time you read it, you revisit those emotions."

So, pick up that pen. And as the poet William Wordsworth said, fill your paper with the breathings of your heart. ■